PROJECT REPORT

EXPERIMENT - 6

Project Statement

Design and develop an Automated Color-Based Product Sorting System for Manufacturing Industries. The system automatically detects and sorts products based on color for industrial automation applications.

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Automated Color-Based Product Sorting System for Manufacturing Industries

Project Overview

The Automated Color-Based Product Sorting System is an industrial automation prototype designed to automatically detect and sort products based on their color. The system uses a TCS3200 Color Sensor to identify the color of products moving on a conveyor prototype. The detected color information is processed by the ESP8266 Node MCU, which controls a servo motor to operate the sorting mechanism. Two LED indicators provide visual status information for the detected product category.

The system reduces manual sorting effort, improves sorting consistency, and demonstrates the application of sensor-based automation in manufacturing industries. The prototype provides a simple, low-cost solution that can be further developed for high-speed industrial product-sorting applications.

Main Objectives :

- To automatically detect product colors using the TCS3200 Color Sensor.
- To process the detected color using the ESP8266 Node MCU.
- To automatically sort products using a servo motor.
- To reduce manual effort and human errors in product sorting.
- To improve sorting efficiency and consistency.
- To demonstrate a low-cost industrial automation solution.

Problem Statement

In manufacturing industries, sorting products based on color is often performed manually, which requires continuous human effort and can lead to errors, inconsistent sorting, and reduced productivity. Manual sorting also becomes difficult when the number of products increases.

Therefore, there is a need for an automated color-based product sorting system that can detect the color of products and automatically separate them into different categories. The proposed system uses a TCS3200 Color Sensor to detect product color, an ESP8266 Node MCU to process the detected color, a servo motor to control the sorting mechanism, and a conveyor prototype to transport the products. This system aims to provide a simple, reliable, and low-cost solution for automated product sorting in manufacturing applications.

Problems Addressed:

- **Manual Sorting:** Reduces the need for manual product sorting based on color.
- **Human Errors:** Minimizes errors during color identification and product classification.
- **Time Consumption:** Reduces the time required for repetitive sorting operations.
- **Inconsistent Sorting:** Provides more consistent sorting compared with manual operations.
- **Low Productivity:** Improves the efficiency of repetitive product-sorting tasks.
- **Continuous Operation:** Enables the sorting process to operate continuously with limited human intervention.
- **Industrial Automation:** Provides a simple and low-cost approach for implementing automated sorting in manufacturing industries.

Proposed Solution

The proposed system is an Automated Color-Based Product Sorting System that automatically identifies and separates products according to their color.

- A conveyor prototype moves the products through the sorting area.
- The TCS3200 Color Sensor detects the color of each product.
- The ESP8266 Node MCU processes the sensor readings and identifies the product category.

- Based on the detected color, the ESP8266 controls the servo motor.
- The servo motor operates a sorting gate to direct the product to the appropriate category.
- Two LED indicators provide visual indication of the detected sorting condition.
- The system reduces manual effort, minimizes sorting errors, and improves the consistency of the sorting process.

Proposed Operation:

- The system is powered ON and the ESP8266 Node MCU initializes all connected components.
- The product is placed on the conveyor prototype.
- The conveyor moves the product toward the color detection area.
- The TCS3200 Color Sensor detects the color of the product.
- The sensor sends the color-related output to the ESP8266 Node MCU.
- The ESP8266 processes the sensor readings and identifies the product color.
- The detected color is compared with the predefined/calibrated color values.
- The controller activates the corresponding LED indicator.
- The servo motor rotates the sorting mechanism according to the detected color.
- The product is directed into its appropriate sorting category.
- The servo returns to its initial position.
- The process repeats automatically for the next product.

System Architecture

The system architecture consists of Input, Processing, Transport, Actuation, and Indication sections.

Input Section:

- The TCS3200 Color Sensor detects the color of the product.
- It produces frequency-based output corresponding to the detected color.

Processing Section:

- The ESP8266 Node MCU receives the sensor output.
- It processes and compares the detected color with calibrated values.
- It decides the required sorting position.

Transport Section:

- The Conveyor Prototype moves the product through the sensing and sorting areas.
- It ensures that the product reaches the color sensor and sorting mechanism.

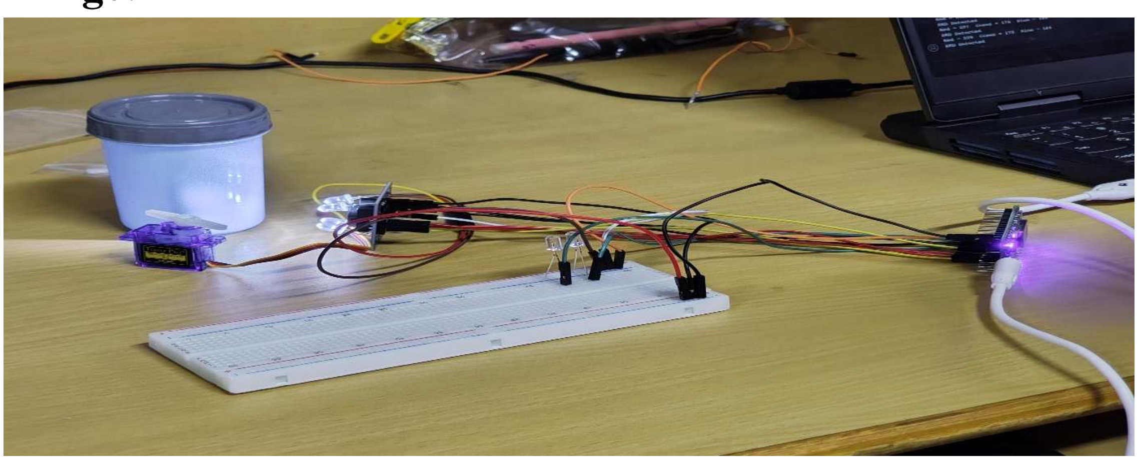
Actuation Section:

- The Servo Motor receives the control signal from the ESP8266.
- It moves the sorting gate according to the detected product color.
- The product is directed to the appropriate category.

Indication Section:

- **LED 1** indicates the first sorting condition.
- **LED 2** indicates the second sorting condition.

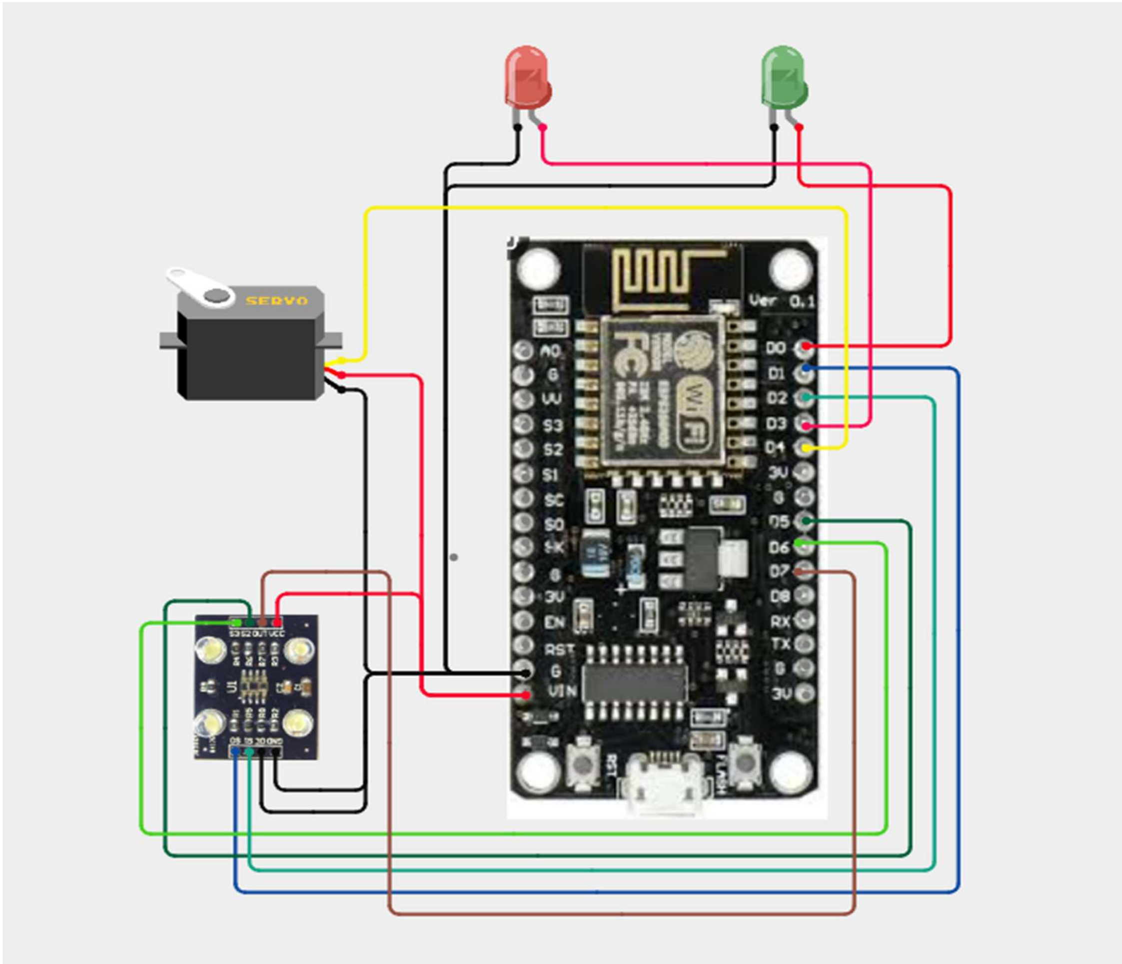
Image:



Circuit Table

S. No.	Component	Pin/Terminal	Connection to ESP8266	Function
1	TCS3200 Color Sensor	S0	D1 (GPIO5)	Frequency scaling control
		S1	D2 (GPIO4)	Frequency scaling control
		S2	D5 (GPIO14)	Color filter selection
		S3	D6 (GPIO12)	Color filter selection
		OUT	D7 (GPIO13)	Color frequency output
		VCC	Appropriate supply	Power supply
		GND	GND	Common ground
2	Servo Motor	Signal	D4 (GPIO2)	Controls sorting mechanism
		VCC	Suitable servo supply	Power supply
		GND	GND	Common ground
3	LED 1	Anode	D3 (GPIO0) through resistor	First color/status indication
		Cathode	GND	Ground connection
4	LED 2	Anode	D8 (GPIO15) through resistor	Second color/status indication
		Cathode	GND	Ground connection
5	Conveyor Prototype	Mechanical section	—	Transports products
6	Breadboard	Power/connection rails	—	Circuit assembly
7	Jumper Wires	—	Between components	Electrical connections

Circuit Design



Important Pin Mapping:

- TCS3200 S0 → ESP8266 D1 (GPIO5) – Frequency scaling
- TCS3200 S1 → ESP8266 D2 (GPIO4) – Frequency scaling
- TCS3200 S2 → ESP8266 D5 (GPIO14) – Color filter selection
- TCS3200 S3 → ESP8266 D6 (GPIO12) – Color filter selection
- TCS3200 OUT → ESP8266 D7 (GPIO13) – Color sensor output
- Servo Signal → ESP8266 D4 (GPIO2) – Sorting mechanism control
- LED 1 → ESP8266 D3 (GPIO0) – First sorting indication
- LED 2 → ESP8266 D8 (GPIO15) – Second sorting indication
- TCS3200 VCC → Appropriate supply – Sensor power
- TCS3200 GND → GND – Common ground
- Servo GND → GND – Common ground

Algorithm

- Start the system.
- Initialize the ESP8266 Node MCU, TCS3200 sensor, servo motor, and LED indicators.
- Configure the TCS3200 frequency scaling and color filter pins.
- Wait for a product to reach the color detection area.
- Detect the product color using the TCS3200 Color Sensor.
- Read the sensor's color-frequency values.
- Send the sensor readings to the ESP8266 Node MCU.
- Compare the readings with the calibrated color values.
- Identify the product's color category.
- Turn ON the corresponding LED indicator.
- Rotate the servo motor to the appropriate sorting position.
- Direct the product into the corresponding sorting category.
- Return the servo motor to its initial position.
- Wait for the next product.
- Repeat the process continuously.
- Stop the system when the power is switched OFF.

Coding

```
#include <Servo.h>


// ------------ TCS3200 Pins -------------

#define S0 5       // D1

#define S1 4       // D2

#define S2 14      // D5

#define S3 12      // D6

#define OUTPIN 13  // D7
```

```
// ------------ Servo & LEDs -------------
#define SERVO_PIN 2      // D4
#define GREEN_LED 16     // D0
#define RED_LED 0        // D3

Servo sorter;

int redValue, greenValue, blueValue;

void setup()
{
  Serial.begin(9600);

  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(OUTPIN, INPUT);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);

  // Frequency Scaling = 20%
  digitalWrite(S0, HIGH);
  digitalWrite(S1, LOW);
```

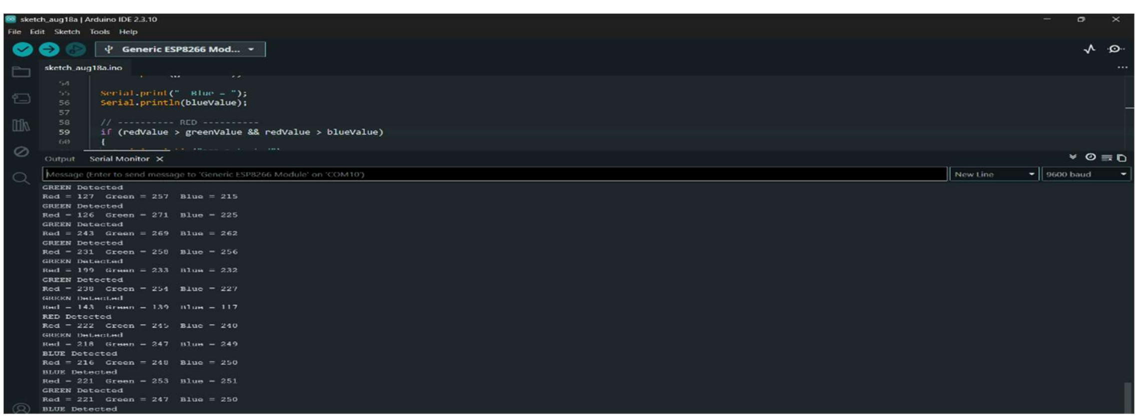
```
  sorter.attach(SERVO_PIN);
  sorter.write(90); // Home Position

  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, LOW);

  Serial.println("Color Sorting System Started");
}

void loop()
{
  readColor();

  Serial.print("Red = ");
  Serial.print(redValue);

  Serial.print("  Green = ");
  Serial.print(greenValue);

  Serial.print("  Blue = ");
  Serial.println(blueValue);

  // ---------- RED ----------
  if (redValue > greenValue && redValue > blueValue)
  {
    Serial.println("RED Detected");
```

```
  digitalWrite(RED_LED, HIGH);
  digitalWrite(GREEN_LED, LOW);

  sorter.write(0);
  delay(2000);
}

// ---------- GREEN ----------
else if (greenValue > redValue && greenValue > blueValue)
{
  Serial.println("GREEN Detected");

  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(RED_LED, LOW);

  sorter.write(90);
  delay(2000);
}

// ---------- BLUE ----------
else if (blueValue > redValue && blueValue > greenValue)
{
  Serial.println("BLUE Detected");

  digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(GREEN_LED, HIGH);

    sorter.write(180);
    delay(2000);
  }

  // ---------- UNKNOWN ----------
  else
  {
    Serial.println("Unknown Color");

    digitalWrite(RED_LED, LOW);
    digitalWrite(GREEN_LED, LOW);
  }

  // Return Servo Home
  sorter.write(90);

  delay(1000);
}

// ========================
// Read RGB Values
// ========================

void readColor()
```

```
{
  // RED
  digitalWrite(S2, LOW);
  digitalWrite(S3, LOW);
  delay(100);
  redValue = pulseIn(OUTPIN, LOW);

  // GREEN
  digitalWrite(S2, HIGH);
  digitalWrite(S3, HIGH);
  delay(100);
  greenValue = pulseIn(OUTPIN, LOW);

  // BLUE
  digitalWrite(S2, LOW);
  digitalWrite(S3, HIGH);
  delay(100);
  blueValue = pulseIn(OUTPIN, LOW);
}
```

Coding Output:



System Working

- The system is powered ON, and the ESP8266 Node MCU initializes the TCS3200 sensor, servo motor, and LED indicators.
- The conveyor prototype moves the product toward the color detection area.
- When the product reaches the sensing position, the TCS3200 Color Sensor detects its color and generates corresponding frequency signals.
- The ESP8266 Node MCU receives and processes the sensor signals to identify the product color.
- The detected color is compared with the predefined/calibrated color values stored in the program.
- Based on the identified color, the corresponding LED indicator is activated.
- The ESP8266 sends a control signal to the servo motor.
- The servo motor rotates the sorting mechanism to the required position.
- As the product moves forward on the conveyor, the sorting mechanism directs it into the appropriate category.
- After sorting, the servo returns to its initial position and the system waits for the next product.
- This process continues automatically, providing continuous color-based product sorting with reduced manual intervention.

Bill of Materials

S. No.	Component	Quantity
1	ESP8266 Node MCU	1
2	TCS3200 Color Sensor	1
3	Servo Motor	1
4	Conveyor Prototype	1
5	LED Indicators	2
6	Breadboard	1
7	Jumper Wires	1 Set

Prototype Implementation

The prototype is implemented by integrating the ESP8266 Node MCU, TCS3200 Color Sensor, servo motor, conveyor prototype, and LED indicators into a single automated sorting system.

- The conveyor prototype is arranged to move products through the color detection and sorting areas.
- The TCS3200 Color Sensor is mounted above the conveyor at a fixed height so that it can detect the color of each product as it passes through the sensing area.
- The ESP8266 Node MCU is connected to the TCS3200 sensor through the breadboard and is programmed to process the color sensor readings.
- The servo motor is installed at the sorting point and connected to a simple mechanical gate or deflector.
- Two LED indicators are connected to the ESP8266 to indicate different detected color categories.
- The TCS3200 sensor is calibrated using sample products with known colors. The obtained sensor readings are used to set suitable color-detection thresholds in the program.
- When a product reaches the sensing area, the sensor detects its color and sends the output to the ESP8266.
- The ESP8266 identifies the product category and controls the servo motor accordingly.
- The servo rotates the sorting gate and directs the product toward the appropriate category.
- After sorting, the servo returns to its initial position, allowing the next product to be processed.

Prototype Flow:

Product → Conveyor → TCS3200 Sensor → ESP8266 Processing → Servo Sorting Gate → Sorted Product

Hardware Setup:

The hardware setup consists of the ESP8266 Node MCU, TCS3200 Color Sensor, servo motor, conveyor prototype, LED indicators, breadboard, and jumper wires.

- **Controller Setup:**Place the ESP8266 Node MCU on the breadboard and use it as the main control unit.
- **Color Sensor Setup:**Mount the TCS3200 Color Sensor above the conveyor path at a fixed distance. Connect its S0, S1, S2, S3, and OUT pins to the ESP8266 according to the pin mapping.
- **Servo Motor Setup:**Fix the servo motor near the end of the conveyor and attach a simple sorting gate or arm to its shaft. Connect the servo signal to the assigned ESP8266 GPIO.
- **LED Setup:**Connect the two LED indicators to the ESP8266 through suitable current-limiting resistors. Use them to indicate different sorting conditions.
- **Conveyor Setup:**Position the conveyor so that products pass directly below the TCS3200 sensor and then reach the servo-operated sorting mechanism.
- **Wiring Setup:**Use jumper wires to connect the sensor, servo, LEDs, and ESP8266. Ensure that the required grounds are connected properly.
- **Programming Setup:**Upload the color detection and sorting program to the ESP8266 using the Arduino IDE.
- **Calibration:**Place sample products of known colors under the TCS3200 sensor and record their readings. Use these readings to set suitable color thresholds in the program.

Prototype Working

The prototype works by automatically detecting the color of products and sorting them into different categories.

- The product is placed on the conveyor prototype.
- The conveyor moves the product toward the TCS3200 Color Sensor.

- The sensor detects the color and generates corresponding frequency signals.
- The ESP8266 Node MCU receives and processes these signals.
- The controller compares the sensor readings with the calibrated color values.
- The corresponding LED indicator is activated.
- The ESP8266 sends a control signal to the servo motor.
- The servo rotates the sorting gate according to the detected color.
- The product is directed to the appropriate sorting category.
- The servo returns to its initial position, and the system is ready for the next product.

Expected Prototype Output

Detected Product Color	**LED Indication**	**Servo Action**	**Expected Output**
Color Category 1	LED 1 ON	Rotates to Position 1	Product directed to Category 1
Color Category 2	LED 2 ON	Rotates to Position 2	Product directed to Category 2
Unknown/Other Color	LEDs OFF	Remains at neutral position	Product follows default path

Prototype Demonstration

For the demonstration, products with different colors are placed individually on the conveyor. As each product moves toward the sensing area, the TCS3200 Color Sensor detects its color. The ESP8266 processes the sensor output and identifies the corresponding color category.

Based on the detected color, the appropriate LED turns ON and the servo motor rotates the sorting gate. The moving product is then directed into the respective category. After the product passes the sorting mechanism, the servo returns to its initial position. The same process is repeated for multiple products to demonstrate automatic and continuous color-based sorting.

Results & Performance Analysis

The developed prototype demonstrates the successful operation of an automated color-based sorting system. The TCS3200 sensor detects the color of the products and provides the required input to the ESP8266. The ESP8266 processes the sensor readings and controls the servo motor according to the detected color.

The prototype can effectively sort products with clearly distinguishable colors when the sensor is properly calibrated. The performance can be affected by ambient lighting, sensor distance, product surface, product alignment, and conveyor speed.

Expected Results:

- Successful detection of predefined product colors.
- Automatic classification of products based on color.
- Automatic operation of the servo sorting mechanism.
- Correct indication through the LED indicators.
- Reduction in manual sorting effort.
- Improved consistency in repetitive sorting operations.
- Successful demonstration of a basic industrial automation concept.
- Reduced possibility of human error during color-based sorting.
- Continuous operation with limited human intervention.

Performance Analysis

Parameter	Expected Performance	Remarks
Color Detection	Good	Depends on sensor calibration
Sorting Accuracy	High for distinct colors	Similar shades may be difficult to distinguish
Servo Response	Fast	Depends on servo movement and program timing
Sorting Consistency	Good	Improved with fixed product position
Human Intervention	Low	Mainly required for loading and monitoring
Repeatability	Good	Stable lighting improves repeatability
Response Time	Low	Depends on sensor reading and servo movement
Major Error Sources	Ambient light and reflections	Controlled lighting improves performance
Automation Level	Automatic	Color detection and sorting are automated
Industrial Applicability	Prototype level	Can be upgraded with industrial components